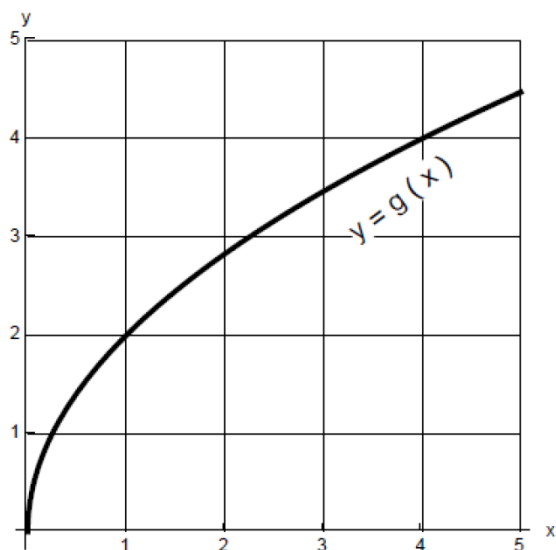


Problem 6

Problem 6

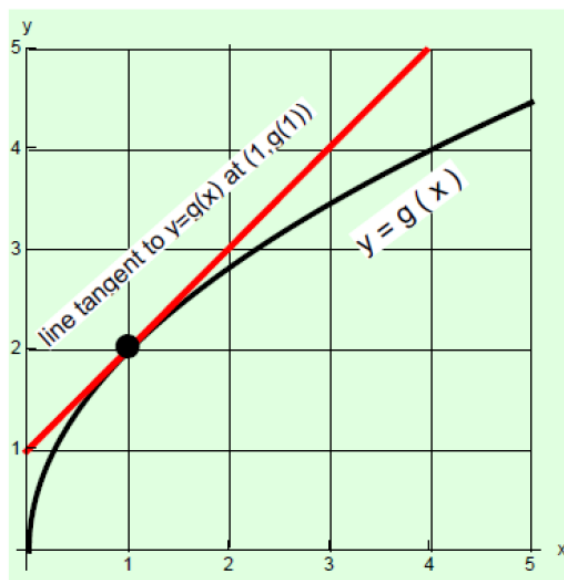
Problem 1 The graph of a function g is given below. Using the graph, estimate the derivative at the given point:

$$\frac{d}{dx} \left(\frac{xg(x)}{x+3} \right) \text{ at } x = 1$$



Explanation. From the graph it appears that $g(1) = 2$. We can draw the line tangent to the curve $y = g(x)$ at the point $(1, g(1))$

Problem 6



Since $g'(1)$ is the slope of the line tangent to the curve $y = g(x)$ at the point $(1, g(1))$, we can estimate it from the figure above. The tangent line seems to pass through the points $(1, 2)$ and $(2, 3)$, so its slope is $\frac{3-2}{2-1} = 1$.

$$\begin{aligned}
 \left[\frac{d}{dx} \left(\frac{xg(x)}{x+3} \right) \right]_{x=1} &= \left[\frac{(g(x) + xg'(x))(x+3) - xg(x)}{(x+3)^2} \right]_{x=1} \\
 &= \frac{(g(1) + 1g'(1))(1+3) - (1)g(1)}{(1+3)^2} \\
 &\approx \frac{(2 + 1(1))(4) - 1(2)}{4^2} \\
 &= \frac{(2+1)(4) - 2}{16} \\
 &= \frac{3(4) - 2}{16} \\
 &= \frac{12 - 2}{16} \\
 &= \frac{10}{16} \\
 &= \frac{5}{8}
 \end{aligned}$$